Serra, ES, Brazil
October 28, 2026

To:
The Editor-in-Chief
Control Engineering Practice
Elsevier / International Federation of Automatic Control (IFAC)

Subject: Submission of Original Research Article titled *"Thermal-Crosstalk-Aware Scheduling and Model Predictive Dynamic Voltage and Frequency Scaling for Spaceborne Multi-Chip AI Clusters in Deep Vacuum"*

Dear Editor-in-Chief,

We are pleased to submit our original research manuscript titled **"Thermal-Crosstalk-Aware Scheduling and Model Predictive Dynamic Voltage and Frequency Scaling for Spaceborne Multi-Chip AI Clusters in Deep Vacuum"** for consideration for publication in *Control Engineering Practice*.

Motivation and Relevance to *Control Engineering Practice*:
Modern Earth Observation satellites increasingly embed commercial off-the-shelf (COTS) multi-core neural accelerators to perform real-time autonomous data triaging in Low Earth Orbit (LEO). In the deep vacuum of space, convective heat transfer is absent ($h \approx 0$), restricting thermal dissipation strictly to solid conduction and highly non-linear fourth-order Stefan-Boltzmann radiation to deep space ($q \propto T^4$). In compact multi-chip architectures, intensive AI workloads induce severe lateral conductive heat diffusion (*thermal crosstalk*), causing localized hot spots and steep spatial temperature gradients that severely degrade conventional terrestrial reactive governors (e.g., Linux ondemand and uncoordinated PID loops).

This article addresses this critical engineering control challenge by developing and evaluating **TCAS-DVFS**:

1. **Practical Control Design:** An Extended Kalman Filter (EKF) using Joseph-stabilized covariance updates estimates silicon junction temperatures under realistic sensor noise ($\sigma_s = 0.25$ K).
2. **Constrained LTV MPC-DVFS Formulation:** Operating frequencies are coordinated online at 0.5 Hz via a discrete linear time-varying Model Predictive Controller that solves a condensed convex quadratic program using a warm-started ADMM splitting iteration paired with an explicit safe fallback policy for non-convergent updates, penalizing excursions above a nominal 83 °C soft thermal cap and actively suppressing inter-die gradients.
3. **Orbital Power Budgeting:** A coupled radiative-conductive power budget explicitly incorporates prospective direct solar flux, Earth albedo, terrestrial infrared, and radiator enthalpy margins under a synthetic analytical orbital forcing model.
4. **Extensive Methodological Verification:** The closed-loop control system is rigorously evaluated over a three-orbit co-simulation (17,004 s, 34,009 steps), achieving zero observed violations of the adopted 85 °C simulation limit, dampening peak and mean spatial gradients by 78.7% and 83.1%, and maintaining competitive inference throughput (124.62 FPS). In addition, a descriptive 30-sample stratified Latin Hypercube Sampling (LHS) sensitivity analysis (±20% perturbation across the four conductive and capacitive network parameters $R_{ij}, R_r, c_i, C_{rad}$) exhibits zero violations (0/30 samples), accompanied by temporal integrator convergence tests, PINN vs. MLP physical benchmarking, and ASTM E1049-85 rainflow validation.

Declarations:

- This manuscript represents original work that has not been published previously and is not currently under consideration for publication elsewhere.
- Its publication has been approved by all authors and tacitly by the responsible authorities at the institution where the work was carried out.
- The authors declare no competing financial or personal conflicts of interest.
- All research code, nominal configurations, ASTM test harnesses, and reproducibility metadata have been released openly on GitHub (`https://github.com/dafnhe-thor-barros/TCAS-DVFS`).

Thank you for your time and consideration of our work. We look forward to hearing from you.

Sincerely,

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